THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 19	Substitute both values.
2	**(c)** $4x + 7y$	Collect like terms separately.
3	**(a)** 23	Brackets first.
4	**(b)** $5x + 7$	Change signs of the subtracted expression.
5	**(b)** 43	Substitute n.
6	**(b)** $3x + 5y$	Multiply price by quantity.
7	**(a)** 15	Square first.
8	**(d)** 30	Substitute both values.
9	**(b)** $8x + 9y$	Collect like terms separately.
10	**(c)** 31	Brackets first.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Expand: $3x + 12$ (1); Add $5x$: $8x + 12$ (1).	1+1
12	$(9x + 2) - (3x - 5)$ (1); $= 6x + 7$ (1).	1+1
13	Substitute: $5^2 + 3(5) - 4$ (1); $= 36$ (1).	1+1
14	Expand: $2x + 14$ (1); Add $6x$: $8x + 14$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 10n + 3$ (1); Difference $= 4n + 7$ (1); At $n = 4$: sum $= 43$, difference $= 23$ (1).	1+1+1
16	For 9: $2(9) + 1 = 19$ (1); Solve $2n + 1 = 29$ (1); $n = 14$ (1).	1+1+1
17	Fare $= 45 + 11k$; $45 + 11k = 133$ gives $11k = 88$ (1); $k = 8$ (1); Fare for 13 km $=$ ₹188 (1).	1+1+1
	OR: Sum $= 14n + 3$ (1); Difference $= 4n + 5$ (1); At $n = 3$: sum $= 45$, difference $= 17$ (1).	1+1+1
18	For 12: $2(12) + 1 = 25$ (1); Solve $2n + 1 = 41$ (1); $n = 20$ (1).	1+1+1
	OR: Fare $= 60 + 9k$; $60 + 9k = 168$ gives $9k = 108$ (1); $k = 12$ (1); Fare for 17 km $=$ ₹213 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Earning $= 48x + 36y$ (1); At given values: $480 + 540$ (1); $=$ ₹1020 (1); Average $= \frac{1020}{25} \approx$ ₹41 (2).	1+1+1+2
	OR: Mother now $= 3r + 4$ (1); Raju after 5 years $= r + 5$ (1); Mother after 5 years $= 3r + 9$ (1); Sum $= 4r + 14$; at $r = 10$: 54 (2).	1+1+1+2
20	Earning $= 52x + 40y$ (1); At given values: $624 + 320$ (1); $=$ ₹944 (1); Average $= \frac{944}{20} \approx$ ₹47 (2).	1+1+1+2
	OR: Mother now $= 3r + 6$ (1); Raju after 3 years $= r + 3$ (1); Mother after 3 years $= 3r + 9$ (1); Sum $= 4r + 12$; at $r = 12$: 60 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $6x + 6$	Change signs of the subtracted expression.
2	**(b)** 73	Substitute n.
3	**(b)** $6x + 2y$	Multiply price by quantity.
4	**(d)** 36	Square first.
5	**(b)** 28	Substitute both values.
6	**(d)** $5x + 11y$	Collect like terms separately.
7	**(a)** 27	Brackets first.
8	**(c)** $7x + 3$	Change signs of the subtracted expression.
9	**(b)** 47	Substitute n.
10	**(c)** $4x + 7y$	Multiply price by quantity.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(11x + 6) - (4x - 1)$ (1); $= 7x + 7$ (1).	1+1
12	Substitute: $6^2 + 2(6) - 9$ (1); $= 39$ (1).	1+1
13	Expand: $4x + 12$ (1); Add $2x$: $6x + 12$ (1).	1+1
14	$(8x + 3) - (2x - 7)$ (1); $= 6x + 10$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 10n + 3$ (1); Difference $= 6n + 9$ (1); At $n = 5$: sum $= 53$, difference $= 39$ (1).	1+1+1
16	For 15: $2(15) + 1 = 31$ (1); Solve $2n + 1 = 51$ (1); $n = 25$ (1).	1+1+1
17	Fare $= 35 + 14k$; $35 + 14k = 119$ gives $14k = 84$ (1); $k = 6$ (1); Fare for 11 km $=$ ₹189 (1).	1+1+1
	OR: Sum $= 16n + -1$ (1); Difference $= 4n + 7$ (1); At $n = 2$: sum $= 31$, difference $= 15$ (1).	1+1+1
18	For 7: $2(7) + 1 = 15$ (1); Solve $2n + 1 = 37$ (1); $n = 18$ (1).	1+1+1
	OR: Fare $= 50 + 13k$; $50 + 13k = 180$ gives $13k = 130$ (1); $k = 10$ (1); Fare for 15 km $=$ ₹245 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Earning $= 45x + 30y$ (1); At given values: $900 + 300$ (1); $=$ ₹1200 (1); Average $= \frac{1200}{30} \approx$ ₹40 (2).	1+1+1+2
	OR: Mother now $= 3r + 2$ (1); Raju after 8 years $= r + 8$ (1); Mother after 8 years $= 3r + 10$ (1); Sum $= 4r + 18$; at $r = 9$: 54 (2).	1+1+1+2
20	Earning $= 60x + 44y$ (1); At given values: $300 + 660$ (1); $=$ ₹960 (1); Average $= \frac{960}{20} \approx$ ₹48 (2).	1+1+1+2
	OR: Mother now $= 3r + 5$ (1); Raju after 6 years $= r + 6$ (1); Mother after 6 years $= 3r + 11$ (1); Sum $= 4r + 17$; at $r = 11$: 61 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Expressions using Letter-Numbers** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 40	Square first.
2	**(c)** 33	Substitute both values.
3	**(c)** $6x + 10y$	Collect like terms separately.
4	**(a)** 21	Brackets first.
5	**(a)** $5x + 9$	Change signs of the subtracted expression.
6	**(c)** 54	Substitute n.
7	**(b)** $8x + 3y$	Multiply price by quantity.
8	**(c)** 60	Square first.
9	**(a)** 11	Substitute then divide.
10	**(c)** 9	Substitute then divide.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Substitute: $4^2 + 5(4) - 7$ (1); $= 29$ (1).	1+1
12	Expand: $5x + 10$ (1); Add $8x$: $13x + 10$ (1).	1+1
13	$(12x + 9) - (5x - 4)$ (1); $= 7x + 13$ (1).	1+1
14	Substitute: $7^2 + 4(7) - 3$ (1); $= 74$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sum $= 17n + 1$ (1); Difference $= 7n + 13$ (1); At $n = 3$: sum $= 52$, difference $= 34$ (1).	1+1+1
16	For 20: $2(20) + 1 = 41$ (1); Solve $2n + 1 = 61$ (1); $n = 30$ (1).	1+1+1
17	Fare $= 40 + 16k$; $40 + 16k = 152$ gives $16k = 112$ (1); $k = 7$ (1); Fare for 12 km $=$ ₹232 (1).	1+1+1
	OR: Sum $= 15n + 1$ (1); Difference $= 7n + 3$ (1); At $n = 6$: sum $= 91$, difference $= 45$ (1).	1+1+1
18	For 11: $2(11) + 1 = 23$ (1); Solve $2n + 1 = 33$ (1); $n = 16$ (1).	1+1+1
	OR: Fare $= 55 + 12k$; $55 + 12k = 163$ gives $12k = 108$ (1); $k = 9$ (1); Fare for 14 km $=$ ₹223 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Earning $= 38x + 28y$ (1); At given values: $608 + 672$ (1); $=$ ₹1280 (1); Average $= \frac{1280}{40} \approx$ ₹32 (2).	1+1+1+2
	OR: Mother now $= 3r + 3$ (1); Raju after 4 years $= r + 4$ (1); Mother after 4 years $= 3r + 7$ (1); Sum $= 4r + 11$; at $r = 14$: 67 (2).	1+1+1+2
20	Earning $= 55x + 35y$ (1); At given values: $440 + 420$ (1); $=$ ₹860 (1); Average $= \frac{860}{20} \approx$ ₹43 (2).	1+1+1+2
	OR: Mother now $= 3r + 7$ (1); Raju after 2 years $= r + 2$ (1); Mother after 2 years $= 3r + 9$ (1); Sum $= 4r + 11$; at $r = 8$: 43 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.